Peak Power: Mock Judge Report

Independent rubric-based evaluation · GridTech Hackathon, Columbia Energy Symposium · prepared 4 Nov 2026

> **Basis and limits.** Scored from repository docs, code, tests and packet outputs (`README.md`, `docs/DATA_GAPS.md`, `docs/PROTOTYPE_INTEGRATION.md`, `tests/README.md`, `data/Symposium_Packet/outputs/*`). The app was not run and no live demo was observed, so demo reliability is unverified.
> **Rubric used:** `docs/Judging Rubric.xlsx`. It has no weights, point values or 1–5 scale. Four judges (Travis, Joe, Hunter, Ryan) each split 100 points across three teams; the three criteria are guiding questions, "not scored separately"; ties are broken by first-place votes. An equal split is 33.3 points per judge (about 133 of 400). Every figure below is an estimated share against **assumed** competitors; nothing is known about Teams 2 and 3.

A. Overall assessment

Strengths: the packet's official checker passes all 384 plans and the recommended default; a rich product (Portfolio, Find a Site, Project plan, Run the plant, Live Operations); What Changed / What to Watch; honest labelling ("Not graded", "Data needed"); candid documentation of limits.

Weaknesses: county-level and energy data are mostly Texas-only; two separate calculation paths; a thin BESS story; breadth that could overwhelm a first-time viewer.

B. Scorecard (the three xlsx questions)

Criterion	Maximum	Estimated (relative)	Confidence
1. Would a developer use this to make a real decision?	None (unscored guide)	Slightly above an equal field, capped by Texas-only energy data	Low–Medium
2. Does the reasoning hold up when challenged?	None (unscored guide)	At or above an equal field; assumptions are disclosed	Medium
3. Does the team deeply understand what it built?	None (unscored guide)	Likely strong; depends on live Q&A	Low
Total	100 per judge, split across 3 teams	**About 31–37 per judge (125–148 of 400)**	Low

Evidence by question

1. **Developer use.** Nine site dimensions across 3,142 counties plus an 87,600-hour hourly dispatch. Cost-optimal default: $283.1M, 62.6% clean, 155 kg CO_2/MWh. Carbon alternative: $399.5M, 78.6% clean, 60.6 kg/MWh. About +$116M (+41%) buys about 95 kg/MWh (`outputs/official_check.json`, `carbon_alternative.json`).

2. **Reasoning under challenge.** Energy grading is a team screen, not a measured dataset (`DATA_GAPS.md` §2). Time to power uses generation-queue waits as a proxy. Dispatch is rule-based, not an optimizer, and battery plans are fixed at 30 MW.
3. **Team understanding.** Docs are unusually candid (two separate engines, unresolved packet label). Live delivery is untested.

Implementation status (from docs and code)

Status	Features
Implemented	Nine variables; 254 Texas and 3,142 US county map with 8 layers; portfolio comparison; BESS V2 rules in the engine; What Changed / Watch (browser `localStorage`); source list; 72-hour live operations
Partial	Nationwide energy (Texas only); weather (separate Explorer; the Prototype only scales 2023 profiles); provenance (sources listed, county source files not in repo)
Not implemented	Real BESS project data; nationwide price, headroom, congestion or reliability; real-world validation

C. Three simulated judges (assumptions, not real judges)

Perspective	Share of 100	Reasoning
A: Technical	28–34	Likes the checker-validated engine and tests. Penalizes proxy grades, rule-based dispatch, split Python/browser engines, and county data that cannot be re-run from source.
B: Energy and industry	27–34	Penalizes missing grid headroom and congestion data, the generation-queue proxy, and the battery-free cost optimum that undercuts the BESS story.
C: Innovation and clarity	34–42	Rewards the combined site-plus-dispatch concept and What Changed / Watch. Risk: breadth can read as busy.

D. Estimated outcome (share of the pot per judge)

Scenario	Per judge	Over 4 judges
Conservative	25–28	About 100–112 of 400
Expected	32–36	About 128–144
Optimistic	40–44	About 160–176

Baseline is an equal field (33.3 per judge). No rank or winning probability is claimed.

E. Top five weaknesses

1. **Confirmed.** Energy is graded only for Texas; 2,888 counties show "Not graded". Price, headroom, congestion and reliability are absent nationwide (`DATA_GAPS.md` §2).
2. **Confirmed.** County data gaps: 62 counties lack water stress, 9 lack hazard, 35 lack a queue region (102 have no queue median); Alaska and Connecticut use outdated geometry; policy is state-level only; county source files are not in the repo.
3. **Confirmed.** BESS uses only the packet catalogue. The cost-optimal portfolio is battery-free. The browser (price threshold) and Python (greedy arbitrage) use different battery heuristics.
4. **Confirmed.** The browser engine and Python API are separate calculation paths (`README.md`). Prototype weather is not live.
5. **Potential.** Demo risk: tabs do not respond until the plan is computed; export and layout planner are untested; the e2e suite takes about 2 minutes; the packet label is unresolved (v6.0.1 vs 6.6.1).

F. Top five improvements

#	Action	Targets	Effort	Possible gain
1	One-page "how a customer answer becomes a recommendation" slide, using the 384-plan library and checker result	Q2, Q3	Low	Highest
2	Center the demo on the cost-vs-carbon alternative ($283M / 155 vs $400M / 61 kg/MWh) and when each is better	Q1	Low	High
3	Pre-warm the plan computation, add a visible "computing" state, rehearse the demo path	Q1	Low	Medium (risk reduction)
4	Load EIA-861 industrial price and reliability by state as separate labelled variables	Q1, Q2	Medium	Medium (only if loaded and verified)
5	Visible provenance and limits panel; state that the battery is not cost-justified under packet rules	Q2	Low	Medium

G. Likely judge questions and suggested answers

1. **Why is the cheapest portfolio battery-free?** The battery's capital cost exceeds the savings from shifting power under packet rules; the lowest-carbon plan uses a 30 MW battery charged from surplus clean energy.
2. **How much does cleaner supply cost?** About +$116M (+41%) for roughly 95 kg/MWh less.
3. **Is the Energy grade measured?** No. It is a labelled team screen for Texas and "Not graded" elsewhere.
4. **Why generation-queue waits for time to power?** A congestion proxy by region, not a load-connection wait.
5. **Can this work outside Texas?** The site map can; the dispatch model covers ERCOT only.
6. **Why rule-based dispatch?** Transparent and checker-validated; a screening tool, not a proven optimum.

7. **How do you show the stress response?** Tier 0 served locally, Tier 1 relocated, Tier 2 paused, each with its cost. Show the actual screen.
8. **How are the nine variables weighted, and why?** Weights were not verified for this report; confirm and prepare a rationale.
9. **How current is the data?** State policy Nov 2026; LBNL queue through 2025; known vintage gaps for Alaska and Connecticut.
10. **What would you change first for a real client?** EIA-861 prices and reliability, utility capacity statements, real BESS project data.

H. Final verdict

Peak Power looks **competitive but unproven**. It likely beats an equal field on honesty and breadth, and loses ground where developer and energy judges push: nationwide energy data, BESS depth and proxy metrics. The largest uncertainties are the other two teams and whether the live demo and Q&A show the depth the docs imply. Items 1–3 above are low effort and can be done before judging.